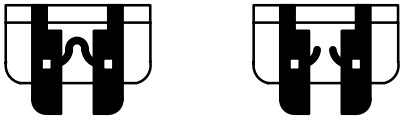
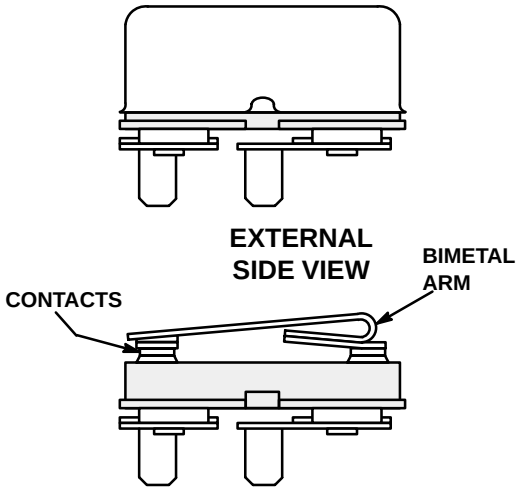
REPLACEMENT OF FUSES/CIRCUIT BREAKERS



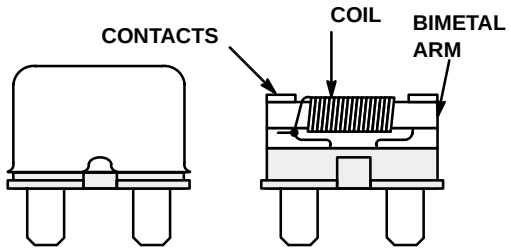
GOOD FUSE BLOWN FUSE

Fuses are mounted in the Engine Compartment Fuse Panel or Interior Fuse Panel. They are identified by the number rating in amperes. Some positions may have either a fuse or a circuit breaker. Be sure to replace a fuse or circuit breaker with the same kind of unit and with the same ampere rating. Remove fuses to check them.

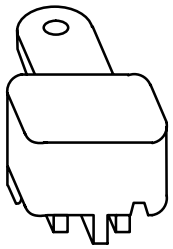
CIRCUIT BREAKER OPERATION



Cycling Fuse Panel Type



Non-Cycling Fuse Panel Type



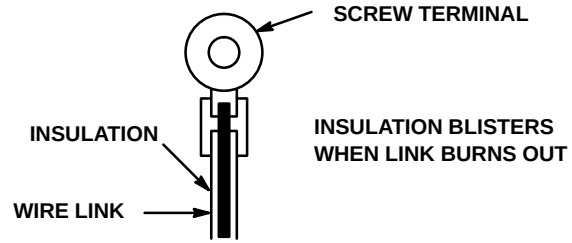
Cycling In-Line Type

Some circuits are protected by circuit breakers (abbreviated “c.b.” in fuse chart). They can be Fuse Panel mounted or in-line. Like fuses, they are rated in amperes.

Each circuit breaker conducts current through an arm made of two types of metal bonded together (bimetal arm). If the arm starts to carry too much current, it heats up. As one metal expands faster than the other, the arm bends, opening the contacts. Current flow is broken. In the cycling type, the arm cools and straightens out. This cycle repeats as long as the overcurrent exists and power is applied.

In the non-cycling type, there is also a coil wrapped around the bimetal arm. When an overcurrent exists and the contacts open, a small current passes through the coil. This current through the coil is not enough to operate a load, but it does heat up both the coil and the bimetal arm. This keeps the arm in the open position until power is removed.

FUSE LINKS



The fuse link is a short length of wire, smaller in gauge than the wire in the protected circuit. The wire is covered with a thick non-flammable insulation. An overload causes the link to heat and the insulation to blister. If the overload remains, the link will melt, causing an open circuit. The links are usually color coded for wire size as follows:

COLOR CODE

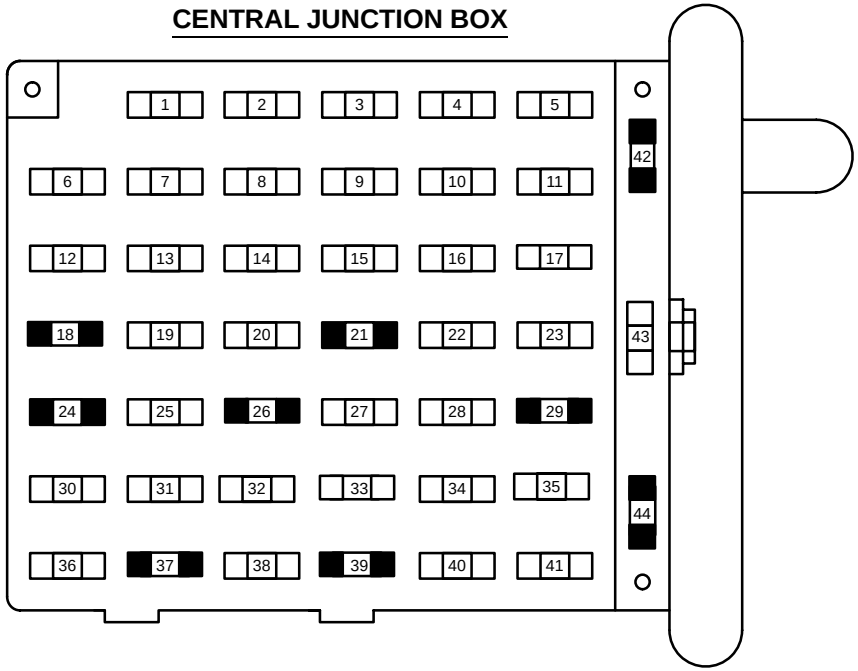
BLUE	20 GA
BROWN OR RED	18 GA
BLACK OR ORANGE	16 GA
GREEN	14 GA

When replacing fuse links, make tight crimp joints or hot solder joints for good connections.

DIODES

Diodes are electrical devices that permit current to flow in one direction only. The current flows in the direction indicated by the arrow.

CENTRAL JUNCTION BOX

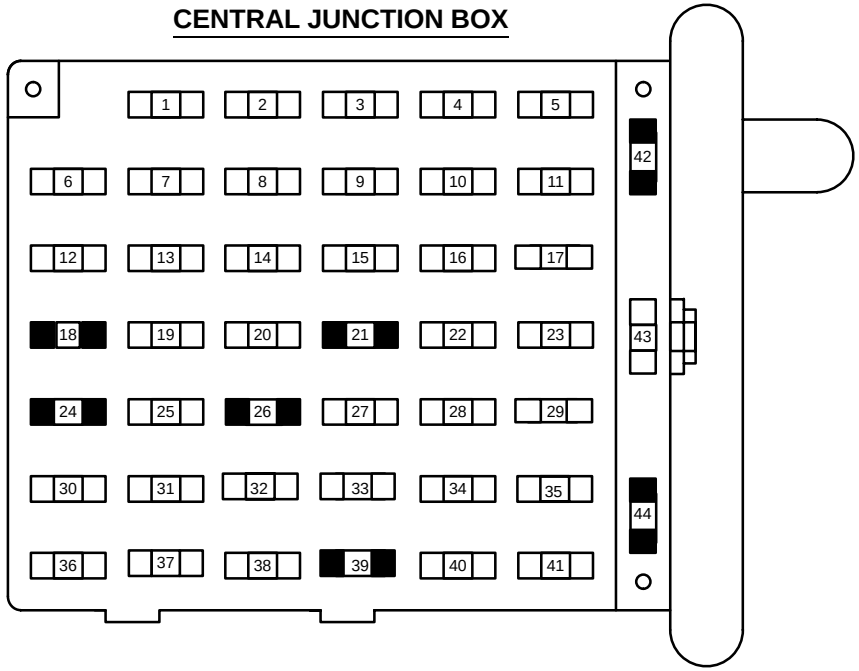


Fuse Value Amps	Color Code
4	Pink
5	Tan
10	Red
15	Light Blue
20	Yellow
25	Natural
30	Light Green

Fuse Position	Amps	Circuits Protected
1	20	4WABS Module
2	15	Instrument Cluster, Warning Chime, 4WABS Relay, Warning Indicators
3	15	Main Light Switch, RKE Module, Radio, Floor Console, Left Video Display, Right Video Display, Overhead Console
4	15	Power Locks w/RKE, Illuminated Entry, Warning Chime, Modified Vehicle, Power Mirrors, Main Light Switch, Courtesy Lamps
5	20	RKE Module, Power Lock Switches, Memory Lock
6	10	Shift Interlock, Speed Control, DRL Module
7	10	Multifunction Switch
8	30	Radio Capacitor(s), Ignition Coil, PCM Diode, PCM Power Relay, Fuel Heater, Glow Plug Relay (Diesel), IDM Relay, Idle Validation Switch
9	30	Wiper Control Module, Windshield Wiper Motor
10	20	Main Light Switch, (External Lamps) Multifunction Switch (Flash to pass)
11	15	Deactivator Switch, Multifunction Switch (Hazards), Brake Pedal Position (BPP) Switch
12	15	Digital Transmission Range (TR) Sensor, Auxiliary Battery Relay
13	15	Selector Switch, Blend Door Actuator
14	5	Instrument Cluster (Air bag and Charge Indicator)
15	5	Trailer Battery Charge Relay
16	30	Power Seats
17	20	Floor Console
18	–	NOT USED
19	10A	Electronic Crash Sensor (ECS) Module
20	5A	Overdrive Cancel Switch
21	–	NOT USED

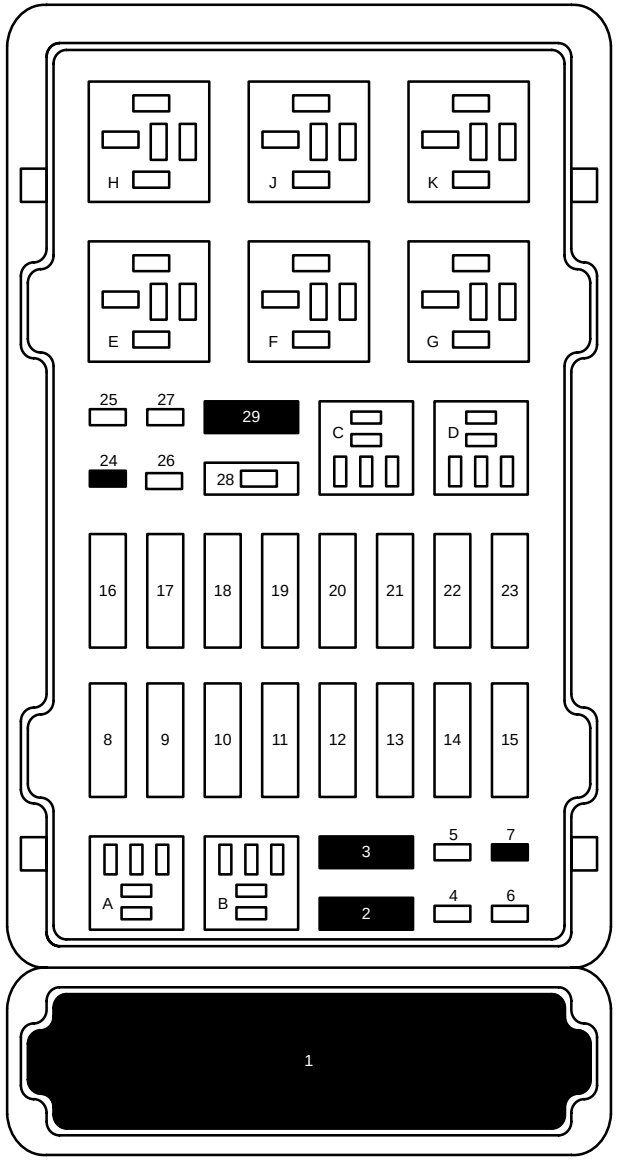
11-3 FUSE PANEL/CIRCUIT PROTECTION

CENTRAL JUNCTION BOX



Fuse Position	Amps	Circuits Protected
22	15	Memory Power Radio
23	20	Cigar Lighter, Data Link Connector (DLC)
24	–	NOT USED
25	10	Left Headlamp (Low Beam)
26	–	NOT USED
27	5	Radio
28	20	Power Plug #1
29	25	Main Light Switch (External Lamps), Multifunction Switch (Flash-to-Pass)
30	15	Headlamps (High Beam Indicator), DRL
31	10	Right Headlamp (Low Beam), DRL
32	5	Power Mirrors
33	20	Floor Console
34	10	Digital Transmission Range (DTR) Sensor
35	30	RKE Module, Auxiliary Blower Motor Relay
36	5	Cluster, Backlite Illumination, Radio, Steering Column Assembly
37	20	Power Plug #2
38	10	Electronic Crash Sensor (ECS) Module
39	–	NOT USED
40	30	Modified Vehicle
41	30	Modified Vehicle
42	–	NOT USED
43	20 c.b.	Power Windows
44	–	NOT USED

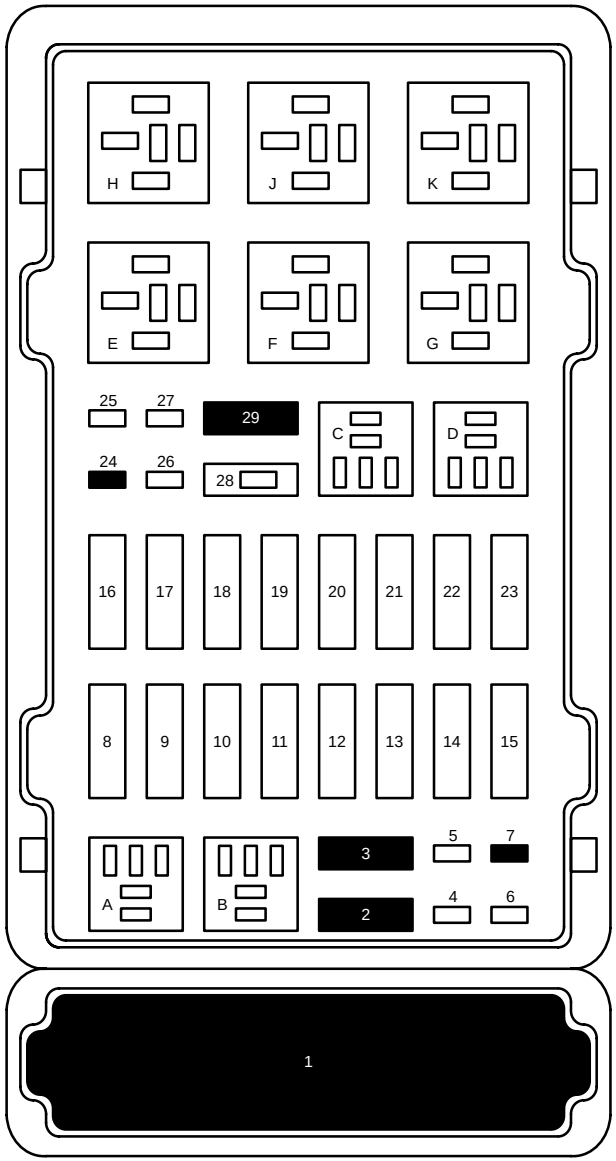
BATTERY JUNCTION BOX



Fuse Position	Amps	Circuits Protected
1	–	NOT USED
2	–	NOT USED
3	–	NOT USED
4	10	PCM Keep Alive Memory, Instrument Cluster
5	10	Right Trailer Turn Signal
6	10	Left Trailer Turn Signal
7	–	NOT USED
8	60	Central Junction Box Fuses 5,11, 23, 38, 4, 10, 16, 22, 28, 32
9	30	PCM Power Relay, Battery Junction Box Fuse 4
10	60	Auxiliary Battery Relay, Engine Compartment Fuses 14, 22
11	30	IDM Relay (Diesel)
12	60	Engine Compartment Fuses 26, 27
13	50	Blower Motor Relay (Blower Motor)
14	30	Trailer Running Lamps Relay, Trailer Reversing Lamps Relay
15	40	Main Light Switch
16	50	RKE Module, Auxiliary Blower Motor Relay
17	30	Fuel Pump Relay
18	60	Central Junction Box Fuses 40, 41
19	60	4WABS Module
20	20	Electric Brake Controller
21	50	Modified Vehicle Power
22	40 (60)	Trailer Battery Charge Relay (Stripped Chassis)
23	60	Ignition Switch
24	–	NOT USED
25	20	NGV Module
26	10	A/C Clutch Relay (4.2L)
27	15	DRL Module, Horn Relay
28	–	PCM Diode
29	–	NOT USED
A	–	4 Wheel Anti-Lock Brake System (4WABS) Relay
B	–	Brake Switch Relay

11-5 FUSE PANEL/CIRCUIT PROTECTION

BATTERY JUNCTION BOX



Fuse Position	Amps	Circuits Protected
C	–	Trailer Reversing Lamps Relay
D	–	Trailer Running Lamps Relay
E	–	Trailer Battery Charge Relay
F	–	IDM Relay (DIESEL), A/C Clutch Relay (4.2L)
G	–	PCM Relay
H	–	Blower Motor Relay
J	–	Horn Relay
K	–	Fuel Pump Relay

High Current Fuse Value Amps	Color Code
30A Plug-in	Green
40A Plug-in	Orange
50A Plug-in	Red
60A Plug-in	Blue

Fuse Value Amps	Color Code
5A	Tan
10A	Red
15A	Light Blue
20A	Yellow